

Quizotic — Complete Technical Guide

Last updated: 8 April 2026 Written for: Mahesh Dhiman (non-developer owner)

This document explains everything about Quizotic — what it is, how it works, where the code lives, what services power it, and how to manage it independently.

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1. What is Quizotic?

Quizotic is a live interactive quiz and presentation platform — think Kahoot + Mentimeter, built for India.

What it does:

- Hosts create quizzes or interactive presentations
- Participants join on their phones by entering a 6-digit code
- Questions appear in real-time, participants answer on their phones
- Live leaderboard, scores, and analytics after each session
- AI can auto-generate quizzes from topics, URLs, or uploaded documents

Live at: <https://quizotic.live>

2. Where Does the Code Live?

The code exists in **three places**. Think of it like a document that has copies:

A. Your MacBook (the original)

```
/Users/mahesh/Claude/CLAUDE ZECTOR/projects/Quizotic/
```

This is where all development happens. Every file, every change starts here. To open it in Finder: open Finder → press Cmd+Shift+G → paste the path above.

To open it in VS Code and actually read the code:

1. Open VS Code
2. File → Open Folder
3. Navigate to: `/Users/mahesh/Claude/CLAUDE ZECTOR/projects/Quizotic/`
4. Click Open — you'll see all files in the left sidebar

B. GitHub (the backup)

Repository: <https://github.com/mdpidx/claude-zector> (private)

Direct link to Quizotic code on GitHub:

```
https://github.com/mdpidx/claude-zector/tree/main/projects/Quizotic
```

Why you didn't see it: The GitHub repo is named `claude-zector` — it's a master repo that holds ALL your projects in one place (Newsletter, CityIndex, Quizotic, etc.). Quizotic is inside the folder `projects/Quizotic/`. You won't find a separate repo called "quizotic" — it's a subfolder.

How to navigate to Quizotic code on GitHub, step by step:

1. Go to <https://github.com/mdpidx/claude-zector>
2. You'll see a list of folders — click on `projects/`
3. Inside `projects/`, click on `Quizotic/`
4. You're now looking at all the Quizotic source code
5. Click `src/` → `app/` to see all the website pages
6. Click any `.tsx` file to read the actual code for that page

What you'll see: Each file shows its complete source code with syntax highlighting. You can see exactly how long each file is, what it does, and the full history of every change ever made.

C. Railway (the live server)

Dashboard: <https://railway.app/dashboard>

Railway is where the website actually runs. It does NOT show you the code — it just runs it. When we run `railway up`, it takes the code from your MacBook, uploads it to Railway, builds it, and starts the server. Users visit `quizotic.live` → Railway serves the pages.

In the Railway dashboard, you'll see two cards:

- **Postgres** — this is your database (stores user data, quizzes, sessions)
- **Quizotic** — this is the web server running your website

The code itself is NOT visible in Railway. Railway is like a factory — it takes your code and runs it, but the code blueprint stays on your Mac and GitHub.

Summary: Three Places, Three Purposes

Where	What's there	How to see it
Your MacBook	The working code — all files	Open VS Code → Open Folder → <code>/Users/mahesh/Claude/CLAUDE ZECTOR/projects/Quizotic/</code>
GitHub	Backup + version history	https://github.com/mdpidx/claude-zector → <code>projects/</code> → <code>Quizotic/</code>

Railway	The running live website	https://railway.app/dashboard → Quizotic → Deployments
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How a change flows:

```

Your MacBook (Claude edits the code)
  |
  v
GitHub (git push – saves a backup copy with history)
  |
  v
Railway (railway up – makes the change live for all users)

```

3. Tech Stack — Plain English

Technology	What it is	Why we use it
Next.js 16	The framework that builds the website	Industry standard for modern websites, handles both the pages users see and the server logic
React 19	The UI library	Makes the website interactive — buttons, forms, animations
Tailwind CSS	Styling system	Makes things look good — colors, spacing, fonts, responsive design
Socket.io	Real-time communication	The magic that makes quizzes "live" — when host shows a question, all participants see it instantly
PostgreSQL	Database	Stores all data — users, quizzes, sessions, payments
Prisma	Database toolkit	Makes it easy to read/write data from the database using JavaScript
NextAuth.js	Login system	Handles Google login, email login, session management
Node.js	Server runtime	The engine that runs your JavaScript code on the server
TypeScript	Programming language	JavaScript with extra safety checks — catches errors before they reach users

4. External Services & APIs

These are the third-party services Quizotic depends on. Think of them as suppliers to a restaurant — each provides a specific ingredient.

4.1 Google OAuth (Login)

Detail	Value
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What it does	Lets users sign in with their Google account
Where configured	Google Cloud Console (console.cloud.google.com)
Used on	Sign-in page (/auth/signin)
Cost	Free
Limit	Unlimited
If it goes down	Users can't log in (no alternative login currently active)

4.2 Neon (Database)

Detail	Value
What it does	Hosts the PostgreSQL database (all user data, quizzes, sessions)
Dashboard	https://console.neon.tech
Used by	Every page that reads or writes data
Cost	Free tier: 0.5 GB storage, 1 compute. Pro: ~\$19/month
Limit	Free tier: 500 MB storage, 190 compute hours/month
If it goes down	Entire website stops working — nothing can load or save

4.3 Railway (Hosting)

Detail	Value
What it does	Runs the website server 24/7
Dashboard	https://railway.app/dashboard
Cost	Pay-as-you-go: ~\$5-15/month depending on traffic
Limit	No hard limit — scales with traffic
If it goes down	Website is completely offline

4.4 Razorpay (Payments — Primary)

Detail	Value
What it does	Handles Pro subscription payments in INR (UPI, cards, netbanking)
Dashboard	https://dashboard.razorpay.com
Used on	Billing page (/host/billing)
Cost	2% per transaction
Limit	No limit
Webhook URL	https://quizotic.live/api/webhooks/razorpay

If it goes down	Users can't upgrade to Pro, but existing Pro users keep working
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4.5 Stripe (Payments — Secondary)

Detail	Value
What it does	Backup payment processor for USD payments
Dashboard	https://dashboard.stripe.com
Cost	2.9% + 30 cents per transaction
Status	Configured but secondary to Razorpay

4.6 OpenRouter (AI Quiz Generation)

Detail	Value
What it does	Powers AI quiz generation — turns topics/documents into quiz questions
Dashboard	https://openrouter.ai
Model used	Google Gemini 2.5 Flash Preview
Used on	Quiz creator (/host/create) — AI tab
Cost	Pay per token (~\$0.001 per quiz generation)
Limit	Based on your OpenRouter balance
If it goes down	AI generation fails, but manual quiz creation still works

4.7 Cloudflare R2 (Image Storage)

Detail	Value
What it does	Stores images uploaded to quiz questions
Dashboard	Cloudflare Dashboard > R2
Cost	Free: 10 GB storage, 10M reads/month
Limit	Very generous free tier
If it goes down	Image uploads fail, but existing images still show (cached)

4.8 PostHog (Analytics)

Detail	Value
What it does	Tracks user behavior — page views, button clicks, feature usage
Dashboard	https://eu.posthog.com (EU region)
Cost	Free: 1M events/month

Limit	1M events/month on free tier
If it goes down	No impact on users — just loses analytics data

4.9 Resend (Email)

Detail	Value
What it does	Sends magic link login emails
Dashboard	https://resend.com/dashboard
Status	DEFERRED — not active yet (waiting for Cloudflare DNS setup)
Cost	Free: 100 emails/day

4.10 Cloudflare (DNS & CDN)

Detail	Value
What it does	Manages quizotic.live domain, security headers, image CDN
Dashboard	https://dash.cloudflare.com
Cost	Free

5. All Website Pages — What They Do

Public Pages (anyone can see)

URL	What it shows	File location
/	Landing page — features, demo, pricing, CTA	src/app/page.tsx
/demo	Interactive demo of quiz components	src/app/demo/page.tsx
/join	Where participants enter the session code	src/app/join/page.tsx
/terms	Terms of service	src/app/terms/page.tsx
/privacy	Privacy policy	src/app/privacy/page.tsx
/r/[code]	Referral link tracker	src/app/r/[code]/page.tsx

Auth Pages (login flow)

URL	What it shows	File location
/auth/signin	Google login + email login form	src/app/auth/signin/page.tsx
/auth/verify	"Check your email" message after magic link	src/app/auth/verify/page.tsx
/auth/error	Error message when login fails	src/app/auth/error/page.tsx
/auth/onboard	First-time user profile setup (role, organization)	src/app/auth/onboard/page.tsx

Host Dashboard (logged-in users)

URL	What it shows	File location
/host	Analytics dashboard (or onboarding for new users)	src/app/host/(dashboard)/page.tsx
/host/quizzes	All saved quizzes — edit, start, delete	src/app/host/(dashboard)/quizzes/page.tsx
/host/presentations	All saved presentations — present, edit, delete	src/app/host/(dashboard)/presentations/page.tsx
/host/sessions	History of all past sessions	src/app/host/(dashboard)/sessions/page.tsx
/host/participants	All participants across sessions	src/app/host/(dashboard)/participants/page.tsx
/host/billing	Subscription, payments, AI usage	src/app/host/(dashboard)/billing/page.tsx
/host/admin	Admin panel (only for admins)	src/app/host/(dashboard)/admin/page.tsx

Creation Pages

URL	What it shows	File location
/host/create	Quiz builder (manual + AI + templates)	src/app/host/create/page.tsx
/host/present/create	Presentation builder	src/app/host/present/create/page.tsx
/host/templates	Quiz template library	src/app/host/templates/page.tsx

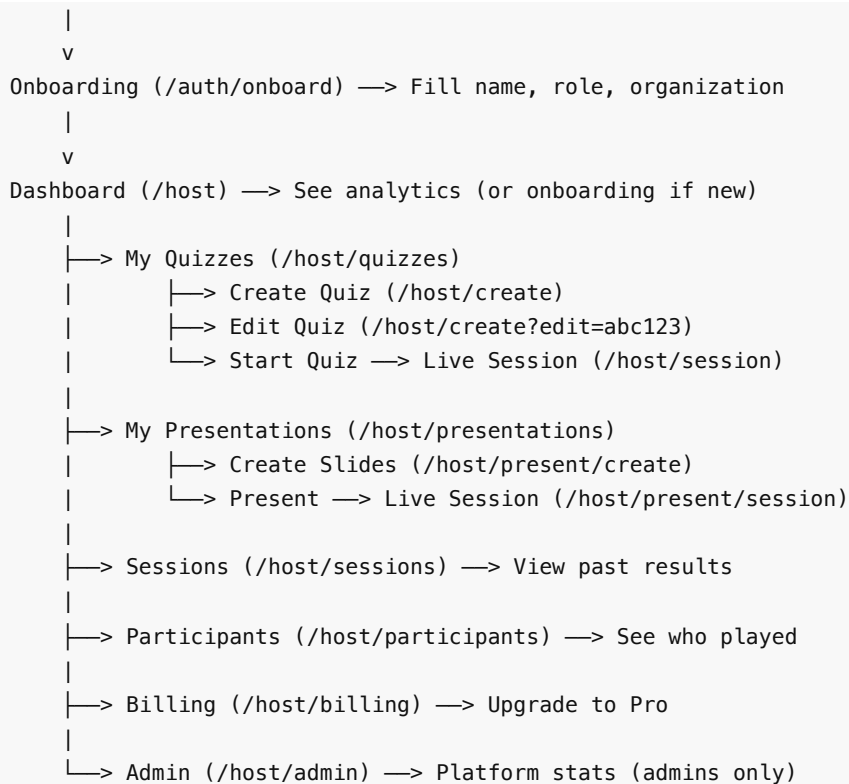
Live Session Pages

URL	What it shows	File location
/host/session	Host's view during a live quiz (controls, leaderboard)	src/app/host/session/page.tsx
/host/present/session	Host's view during a live presentation	src/app/host/present/session/page.tsx

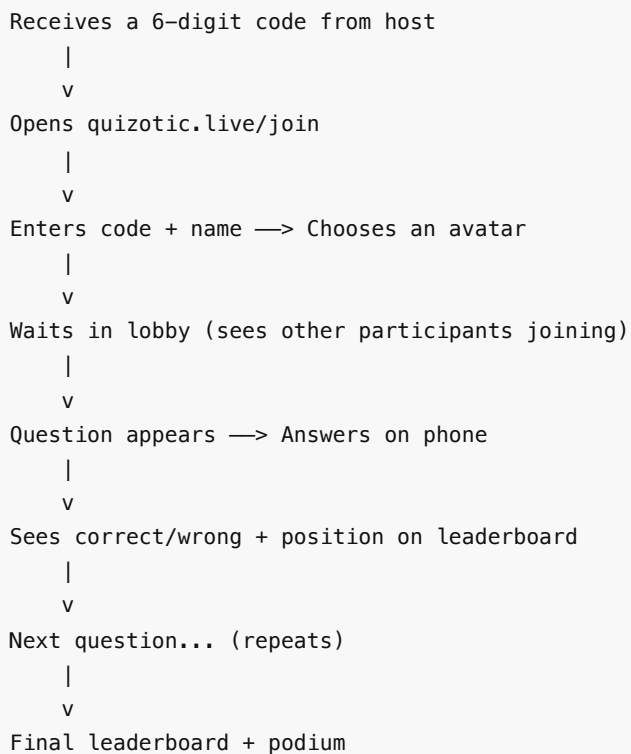
6. How Pages Connect to Each Other

User Journey — Host





User Journey — Participant



Which Pages Call Which APIs

Page	APIs it uses
Dashboard (/host)	GET /api/analytics
My Quizzes	GET /api/quizzes, DELETE /api/quizzes/[id]
My Presentations	GET /api/presentations, DELETE /api/presentations/[id]
Sessions	GET /api/sessions
Participants	GET /api/analytics/participants
Billing	GET /api/billing/status, GET /api/billing/payments, POST /api/billing/checkout
Quiz Creator	POST /api/generate-quiz, POST /api/quizzes, POST /api/upload-image
Slide Creator	POST /api/presentations, POST /api/upload-image
Admin	GET /api/admin/stats, POST /api/admin/grant-pro
Live Quiz Session	WebSocket (Socket.io) — not HTTP API
Join Page	WebSocket (Socket.io) — not HTTP API

7. Database — What Data is Stored

The database has 11 tables. Here's what each stores:

Table	What it stores	Example
User	Every registered user	Name, email, Google profile, role, organization, referral code
Account	Google login details	OAuth tokens, provider IDs
Session	Login sessions	Who's currently logged in
VerificationToken	Magic link tokens	Email verification codes
Subscription	Pro plan status	Plan type, start/end dates, payment provider
Payment	Payment receipts	Amount, currency, invoice URL, status
UsageLog	AI generation tracking	How many AI questions a user has generated
Quiz	Saved quizzes	Title, subject, all questions (as JSON)
Presentation	Saved presentations	Title, all slides (as JSON)
GameSession	Live session results	6-digit code, participants, scores, leaderboard
Referral	Referral tracking	Who referred whom, reward status

Key Relationships

- One **User** can have many **Quizzes**, **Presentations**, **GameSessions**

- One **User** has one **Subscription**
- One **GameSession** links to one **Quiz** or one **Presentation**
- **Referrals** link two **Users** (referrer and referee)

8. Real-Time Quiz Engine (Socket.io)

This is the most complex part of Quizotic. Socket.io enables "instant" communication between the host and all participants.

How it works (simplified)

Host's browser <—WebSocket—> Server (server.mjs) <—WebSocket—> Participant's phone

Unlike normal web pages where you click and wait for a response, WebSocket keeps a permanent connection open. When the host clicks "Next Question", the server instantly pushes that question to every participant's phone.

The Server File

File: `server.mjs` (788 lines — the heart of the real-time system) **Location:**
`/Users/mahesh/Claude/CLAUDE ZECTOR/projects/Quizotic/server.mjs`

This is a custom Node.js server that runs both:

1. The Next.js website (pages, APIs)
2. The Socket.io real-time engine

This is critical: Socket.io CANNOT run on normal hosting (like Vercel). It needs a persistent server, which is why Quizotic runs on Railway with a custom server.

What happens during a live quiz

1. **Host creates session** → Server generates 6-digit code, stores session in memory
2. **Participant joins** → Enters code, server adds them to the session room
3. **Host starts quiz** → Server sends first question to all participants
4. **Participant answers** → Server records answer, sends confirmation
5. **Time runs out** → Server calculates scores, sends leaderboard update
6. **Host clicks next** → Server sends next question
7. **Quiz ends** → Server saves all results to database, sends final leaderboard

In-Memory Storage (Important Limitation)

Active quiz sessions are stored in server memory, NOT in the database. This means:

- **Max 500 concurrent sessions** (configurable)
- If Railway restarts the server, all active sessions are lost
- Completed session results ARE saved to the database permanently

9. Free vs Pro Plan Limits

Feature	Free Plan	Pro Plan
Participants per session	50	Unlimited

AI questions per month	30	750
Questions per AI generation	10	25
Saved quizzes	5	Unlimited
Saved presentations	3	Unlimited
Slides per presentation	10	Unlimited
Session history	3	50
Image uploads per month	20	500
CSV export	No	Yes
PDF export	No	Yes
Spaced retrieval	No	Yes
Quizotic branding	Shown	Hidden

Enforcement file: `src/lib/limits.ts`

10. All Environment Variables (Secrets)

These are the secret keys and passwords that make everything work. They're stored in:

```
/Users/mahesh/Claude/CLAUDE_ZECTOR/secrets/env/quizotic.env
```

The Quizotic project reads this file via a symlink: `.env` -> `../../secrets/env/quizotic.env`

Complete List

Variable	Service	What it is
PORT	Server	Which port the server runs on (4000)
NODE_ENV	Server	"development" or "production"
DATABASE_URL	Neon	Database connection URL
AUTH_SECRET	NextAuth	Secret key for encrypting login tokens
NEXTAUTH_SECRET	NextAuth	Same as AUTH_SECRET
NEXTAUTH_URL	NextAuth	Base URL of the website
GOOGLE_CLIENT_ID	Google	OAuth app ID
GOOGLE_CLIENT_SECRET	Google	OAuth app secret
RAZORPAY_KEY_ID	Razorpay	Payment API key
RAZORPAY_KEY_SECRET	Razorpay	Payment API secret

RAZORPAY_WEBHOOK_SECRET	Razorpay	Webhook verification secret
RAZORPAY_PLAN_MONTHLY	Razorpay	Monthly subscription plan ID
RAZORPAY_PLAN_YEARLY	Razorpay	Yearly subscription plan ID
STRIPE_SECRET_KEY	Stripe	Payment API secret
STRIPE_WEBHOOK_SECRET	Stripe	Webhook verification secret
STRIPE_PRICE_MONTHLY	Stripe	Monthly price ID
STRIPE_PRICE_YEARLY	Stripe	Yearly price ID
RESEND_API_KEY	Resend	Email API key (not active yet)
EMAIL_FROM	Resend	Sender email address
NEXT_PUBLIC_POSTHOG_KEY	PostHog	Analytics public key
NEXT_PUBLIC_POSTHOG_HOST	PostHog	Analytics server URL
R2_ACCOUNT_ID	Cloudflare	R2 storage account
R2_ACCESS_KEY_ID	Cloudflare	R2 API key
R2_SECRET_ACCESS_KEY	Cloudflare	R2 API secret
R2_BUCKET_NAME	Cloudflare	R2 bucket name
R2_PUBLIC_URL	Cloudflare	Public URL for uploaded images
OPENROUTER_API_KEY	OpenRouter	AI API key
QUIZ_AI_MODEL	OpenRouter	Which AI model to use
ADMIN_EMAILS	App	Comma-separated admin email addresses
HOST_DOMAIN	Socket.io	Production host domain (CORS)
JOIN_DOMAIN	Socket.io	Production join domain (CORS)

Where to find/change these

1. **Railway dashboard** → Your project → Variables tab (production values)
2. **Local file** → `/Users/mahesh/Claude/CLAUDE_ZECTOR/secrets/env/quizotic.env` (development values)
3. **Each service's dashboard** → To get new keys if compromised

11. File Structure — Where Everything Is

The entire Quizotic codebase is at:

```
/Users/mahesh/Claude/CLAUDE_ZECTOR/projects/Quizotic/
```

On GitHub, the same folder is at:

<https://github.com/mdpidx/claude-zector/tree/main/projects/Quizotic>

Scale of the codebase: ~12,000 lines of code across ~60 files. The 5 largest files do most of the heavy lifting.

The 5 Most Important Files (Start Here)

File	Lines	What it does	View on GitHub
server.mjs	~800	The real-time quiz engine — the heart of the system	projects/Quizotic/server.mjs
src/app/host/session/page.tsx	~800	Host's live quiz control page	src/app/host/session/page.tsx
src/app/join/page.tsx	~950	Everything a participant sees during a quiz	src/app/join/page.tsx
src/app/host/create/page.tsx	~900	The quiz builder (all tabs — manual, AI, templates)	src/app/host/create/page.tsx
src/app/page.tsx	~700	The public landing page	src/app/page.tsx

Top-Level Files

File	Lines	What it does
server.mjs	~800	The custom server — runs Next.js + Socket.io together. This is what Railway runs.
package.json	~60	Lists all 35 dependencies and run scripts
next.config.ts	~40	Next.js configuration (image domains, security headers)
railway.json	~10	Railway deployment settings
Dockerfile	~20	Instructions for Railway to build the project
prisma/schema.prisma	~120	Database blueprint — defines all 11 tables
.env	—	Secrets file (symlinked to secrets vault — not in GitHub)
tsconfig.json	~20	TypeScript configuration

All Pages (src/app/) — Complete File List

Public Pages (anyone can access)

File	Lines	What users see
src/app/page.tsx	~700	Landing page — features, pricing, sign-up CTA
src/app/join/page.tsx	~950	Participant quiz experience — code entry, avatar, questions, results

src/app/demo/page.tsx	~100	Interactive component demo
src/app/terms/page.tsx	~80	Terms of service
src/app/privacy/page.tsx	~80	Privacy policy
src/app/r/[code]/route.ts	~30	Referral link tracker (redirects to home)

Auth Pages

File	Lines	What users see
src/app/auth/signin/page.tsx	~150	Google login + email login form
src/app/auth/verify/page.tsx	~40	"Check your email" message
src/app/auth/error/page.tsx	~40	Login error message
src/app/auth/onboard/page.tsx	~200	First-time user setup (name, role, org)

Host Dashboard

File	Lines	What hosts see
src/app/host/(dashboard)/page.tsx	~350	Analytics overview (or onboarding for new users)
src/app/host/(dashboard)/quizzes/page.tsx	~250	List of all saved quizzes
src/app/host/(dashboard)/presentations/page.tsx	~240	List of all saved presentations
src/app/host/(dashboard)/sessions/page.tsx	~280	History of all past quiz sessions
src/app/host/(dashboard)/participants/page.tsx	~250	All participants across all sessions
src/app/host/(dashboard)/billing/page.tsx	~400	Subscription, payment history, plan usage
src/app/host/(dashboard)/admin/page.tsx	~300	Admin-only: user stats, grant Pro access

Creation Pages

File	Lines	What hosts do here
src/app/host/create/page.tsx	~900	Full quiz builder (manual + AI + templates + image upload)
src/app/host/present/create/page.tsx	~700	Presentation slide builder
src/app/host/templates/page.tsx	~200	Browse and use built-in quiz templates

Live Session Pages

File	Lines	What happens here
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src/app/host/session/page.tsx	~800	Host controls during a live quiz (timer, votes, leaderboard)
src/app/host/present/session/page.tsx	~500	Host controls during a live presentation

API Routes (`src/app/api/`) — 27 Backend Endpoints

These are NOT pages — they're the server functions that pages call to fetch or save data.

Folder	What it handles
api/auth/[...nextauth]/	Login, logout, session management
api/quizzes/	Save, fetch, delete quizzes
api/presentations/	Save, fetch, delete presentations
api/sessions/	Fetch session history and results
api/analytics/	Dashboard stats, participant data
api/billing/status/	Check if user is Pro or Free
api/billing/payments/	List payment history
api/billing/checkout/	Create Razorpay/Stripe payment link
api/admin/stats/	Platform-wide stats (admin only)
api/admin/grant-pro/	Manually grant Pro to a user
api/user/profile/	Read/update user profile
api/user/referrals/	Referral tracking and rewards
api/generate-quiz/	AI quiz generation via OpenRouter
api/translate-quiz/	Translate quiz to another language
api/upload-image/	Upload images to Cloudflare R2
api/webhooks/razorpay/	Handle Razorpay payment confirmations
api/webhooks/stripe/	Handle Stripe payment confirmations

Reusable Components (`src/components/`)

File	What it does
CircularTimer.tsx	The countdown circle shown during questions
Podium.tsx	Top-3 podium animation at quiz end
SessionReport.tsx	Full results breakdown after a session
ReflectionInsights.tsx	AI-generated debrief after a session

Avatar.tsx	Generates participant avatar images
ImageUpload.tsx	Drag-and-drop image uploader
landing/ (folder)	13 sections of the landing page, each in its own file
host/HostSidebar.tsx	Left sidebar navigation in the host dashboard

Shared Logic ([src/lib/](#))

File	What it does
auth.ts	Login configuration (Google OAuth, magic links)
prisma.ts	Database connection client
billing.ts	Pro/Free plan checking logic
limits.ts	All plan limits in one place (change here to update everywhere)
quiz-types.ts	Data types for quizzes, questions, options
quiz-templates.ts	10+ built-in quiz templates
razorpay.ts	Razorpay payment client
stripe.ts	Stripe payment client
referral.ts	Referral code generation
archetypes.ts	Participant avatar personality types
sounds.ts	Quiz sound effect URLs

Documentation ([docs/](#))

File	What it contains
docs/PRD.md	Product Requirements Document — the original spec
docs/brainstorm.md	Initial brainstorm and feature ideas
docs/feature-comparison.md	Quizotic vs Kahoot vs Mentimeter comparison
docs/cost-analysis.md	Detailed cost breakdown
docs/TECHNICAL-GUIDE.md	THIS document
docs/TECHNICAL-GUIDE.pdf	PDF version of this document

12. How to Deploy Changes

When code changes are made and need to go live:

Step 1: Commit the changes (save to history)

```
cd "/Users/mahesh/Claude/CLAUDE ZECTOR/projects/Quizotic"
git add -A
git commit -m "description of what changed"
```

Step 2: Push to GitHub (backup)

```
git push
```

Step 3: Deploy to Railway (make it live)

```
cd "/Users/mahesh/Claude/CLAUDE ZECTOR/projects/Quizotic"
railway up --detach
```

This uploads the code to Railway, which then:

1. Installs all dependencies
2. Builds the website
3. Starts the server
4. Makes it live at quizotic.live

Build time: Usually 2-4 minutes.

Monitoring the deployment

- Go to <https://railway.app/dashboard>
- Click on your Quizotic project
- You'll see build logs in real-time
- Green = success, Red = something broke

13. How to Make Edits Without Claude

Option A: Use Claude Code (Recommended)

Even without this specific conversation, you can start a new Claude Code session:

1. Open Terminal
2. Navigate to the project: `cd "/Users/mahesh/Claude/CLAUDE ZECTOR/projects/Quizotic"`
3. Run `claude` to start Claude Code
4. Tell it what you want to change
5. It will read the code, make changes, and you approve

Option B: Use VS Code (for small text edits)

1. Open VS Code
2. Open the folder: `/Users/mahesh/Claude/CLAUDE ZECTOR/projects/Quizotic/`
3. Navigate to the file you want to edit
4. Make changes
5. Deploy using the terminal commands from Section 12

Option C: Edit directly on Railway

Railway has an online editor, but it's very limited. Not recommended for code changes.

Option D: Hire a developer

If you need changes beyond what Claude can do, share:

- This document
- The GitHub repo link (github.com/mdpikk/claude-zector)
- Access to Railway, Neon, and other service dashboards

Any Next.js developer can understand and modify this codebase.

Running the website locally (for testing before deploying)

```
cd "/Users/mahesh/Claude/CLAUDE ZECTOR/projects/Quizotic"
npm run dev
```

Then open <http://localhost:4000> in your browser. Changes you make to files will show up instantly (hot reload).

14. Monthly Costs & Service Limits

Current Monthly Cost Estimate

Service	Plan	Cost	Usage
Railway	Pay-as-you-go	~\$5-15/mo	Depends on traffic
Neon (Database)	Free	\$0	500 MB storage
Google OAuth	Free	\$0	Unlimited
Cloudflare	Free	\$0	DNS + R2 storage
PostHog	Free	\$0	1M events/month
OpenRouter	Pay-per-use	~\$1-5/mo	Depends on AI usage
Razorpay	Per transaction	2% of payments	Only when users pay
Resend	Free	\$0	Not active yet
TOTAL		~\$6-20/mo	

Service Limits to Watch

Service	Limit	What happens when exceeded
Neon (Free)	500 MB database	Need to upgrade (\$19/mo) or clean old data
PostHog (Free)	1M events/month	Stops tracking, no user impact

OpenRouter	Balance runs out	AI quiz generation stops working
Cloudflare R2	10 GB storage	Need to upgrade or clean old images
Railway	\$5 included credit	Billing starts after credit used

15. Emergency Troubleshooting

"Website is down"

1. Go to <https://railway.app/dashboard>
2. Check if the server is running (green status)
3. Click on the service → Deployments tab → check for errors
4. If it's crashing, check the logs for error messages

"Users can't log in"

1. Check Google Cloud Console → your OAuth app → is it still active?
2. Check Railway → environment variables → are `GOOGLE_CLIENT_ID` and `GOOGLE_CLIENT_SECRET` set?
3. Check the `NEXTAUTH_URL` matches `https://quizotic.live`

"AI generation not working"

1. Check OpenRouter dashboard → is there balance remaining?
2. Check Railway env vars → is `OPENROUTER_API_KEY` set?

"Payments not processing"

1. Check Razorpay dashboard → are there any failed webhooks?
2. Verify the webhook URL is `https://quizotic.live/api/webhooks/razorpay`
3. Check Railway env vars → are all Razorpay variables set?

"Database is full"

1. Go to Neon console → check storage usage
2. Old `GameSession` records with large `results` JSON are the biggest space users
3. You can delete old sessions via the Neon SQL editor:

```
DELETE FROM "GameSession" WHERE "endedAt" < NOW() - INTERVAL '6 months';
```

Need to change admin emails

Update the `ADMIN_EMAILS` environment variable in Railway → Variables tab. Comma-separated list of email addresses.

Quick Reference Card

What	Where
Live website	https://quizotic.live
Code on your Mac	<code>/Users/mahesh/Claude/CLAUDE_ZECTOR/projects/Quizotic/</code>

GitHub backup	https://github.com/mdpidx/claude-zector (private)
Railway dashboard	https://railway.app/dashboard
Database dashboard	https://console.neon.tech
Google OAuth	https://console.cloud.google.com
Razorpay dashboard	https://dashboard.razorpay.com
PostHog analytics	https://eu.posthog.com
OpenRouter (AI)	https://openrouter.ai
Cloudflare (DNS/CDN)	https://dash.cloudflare.com
Secrets file	/Users/mahesh/Claude/CLAUDE ZECTOR/secrets/env/quizotic.env
Deploy command	cd projects/Quizotic && railway up --detach
Run locally	cd projects/Quizotic && npm run dev → localhost:4000

This document was auto-generated by Claude Code on 7 April 2026. Update it whenever major changes are made to the architecture.